

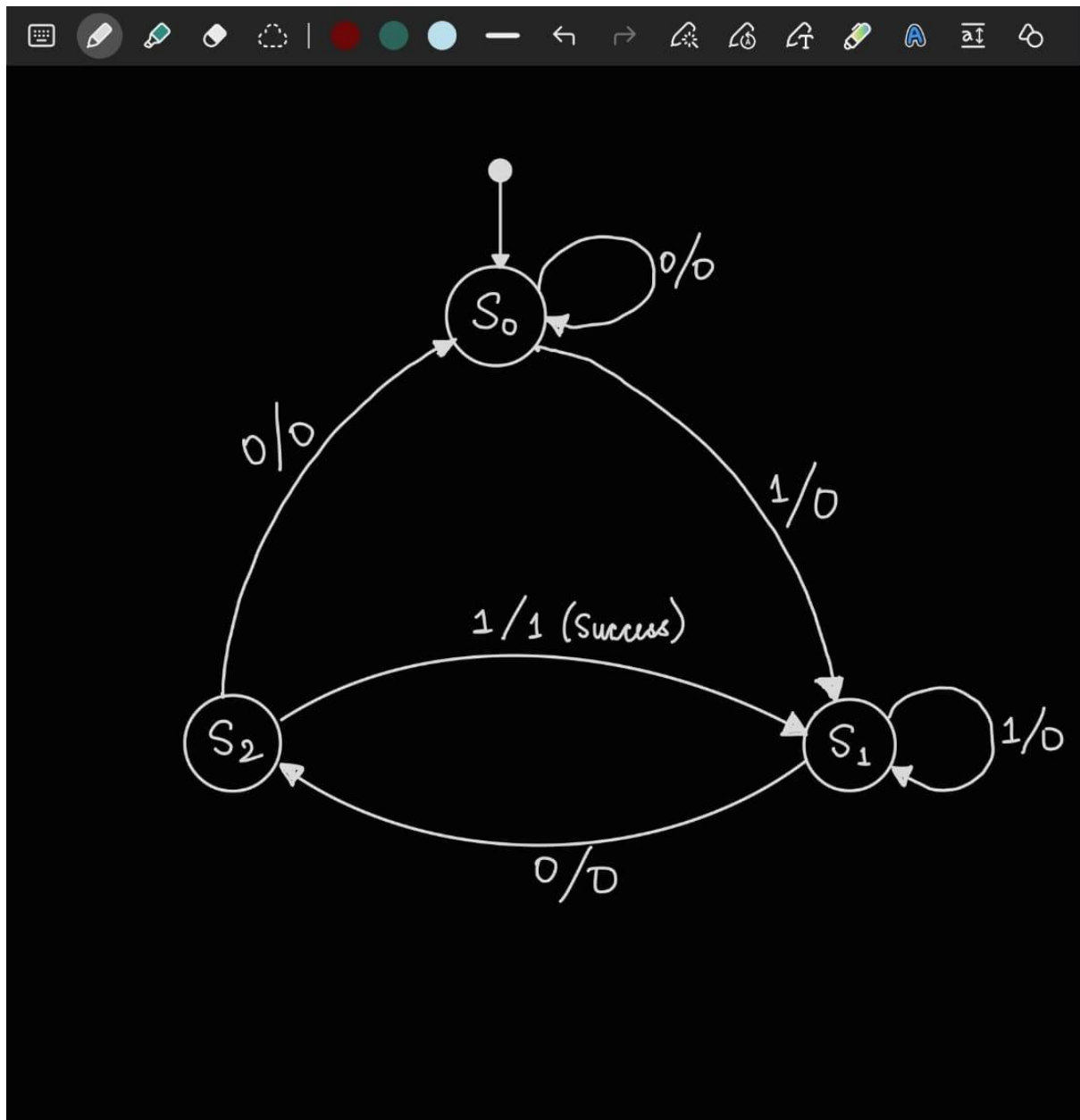
Challenge 2: FSM State Diagrams and Specifications

1. Part A: 101 Sequence Detector (Mealy Machine)

This Finite State Machine detects the overlapping sequence "101". Because it is a Mealy machine, the output is dependent on both the current state and the current input. The output spikes high on the exact transition where the final '1' is received.

State Specifications

State	Meaning
S ₀	Reset / Idle State. The machine is waiting for the sequence to start. It has not detected any relevant bits yet. If a 0 is received, it stays here.
S ₁	Got '1'. The machine has detected the first 1 of the sequence. If another 1 is received, it stays here (acting as a new first 1).
S ₂	Got '10'. The machine has detected 1 followed by 0. It is waiting for the final 1. If a 0 is received, the sequence is broken, and it returns to S ₀ .

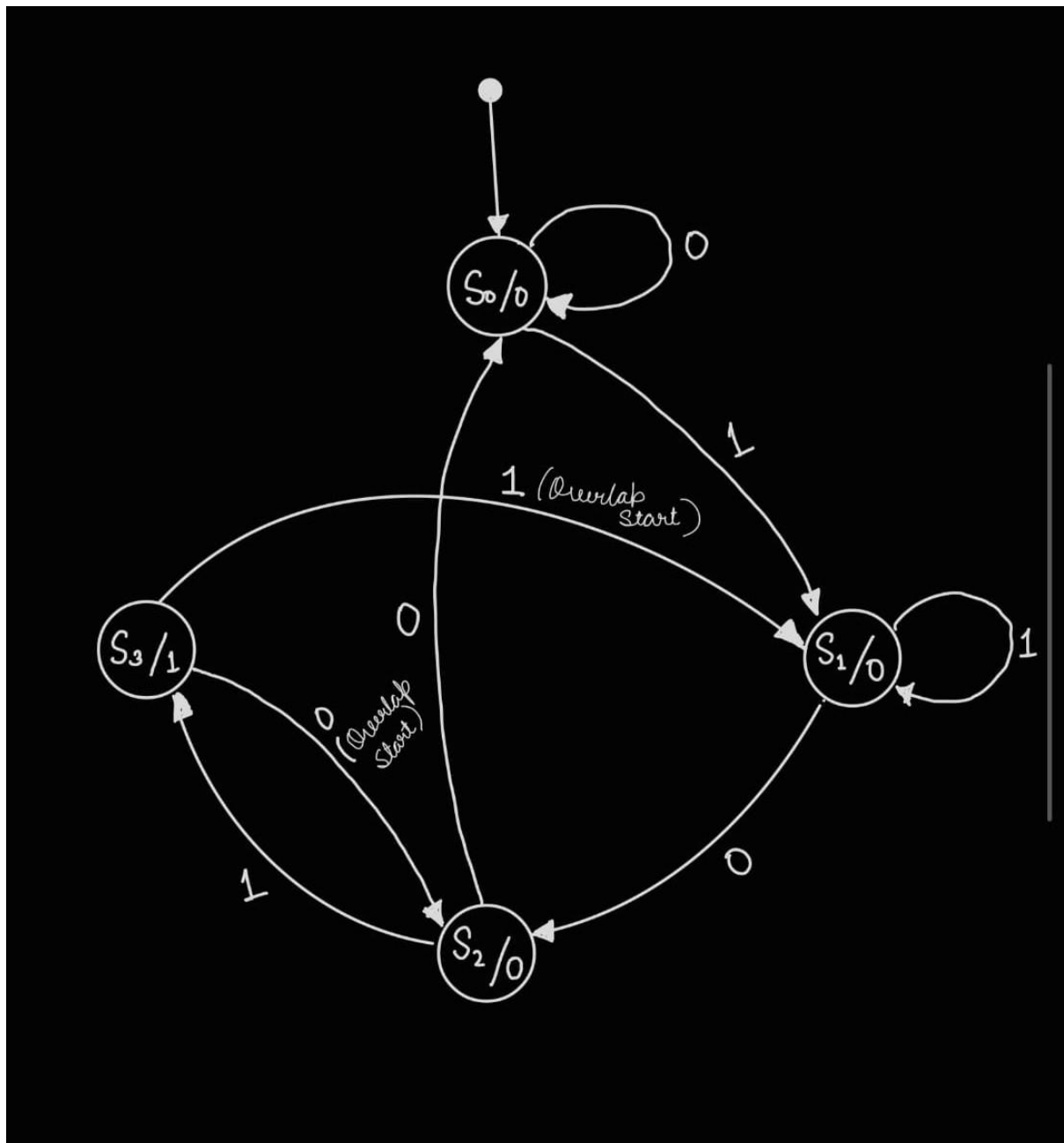


2. Part A: 101 Sequence Detector (Moore Machine)

This Finite State Machine also detects the overlapping sequence "101", but as a Moore machine, the output depends *only* on the current state. Therefore, it requires a 4th state to represent the successful detection phase.

State Specifications

State	Meaning	Output
S_0	Reset / Idle State. Waiting for the first '1'.	0
S_1	Got '1'. The first bit of the sequence is recognized.	0
S_2	Got '10'. The first two bits are recognized.	0
S_3	Got '101' (Success). The full sequence is captured.	1

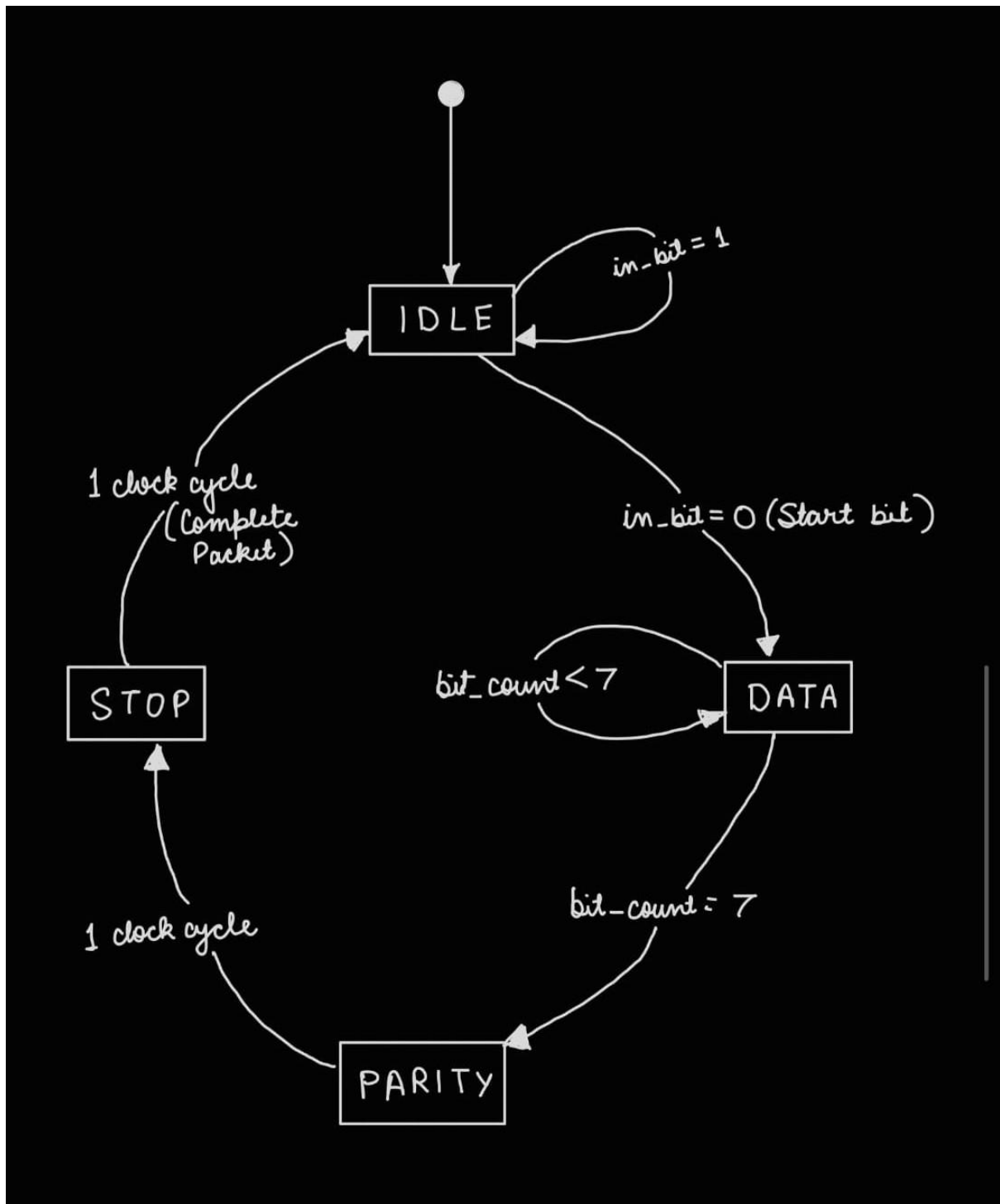


3. Part B: Frame Packet Receiver

This behavioral FSM acts as the control path for a UART-style serial receiver. It manages a datapath to capture an 8-bit payload, verify even parity, and check the stop bit.

State Specifications

State	Meaning	Transition Condition
IDLE	Waiting for Start Bit. The FSM monitors the resting line (1). The internal bit counter is held at 0.	Jumps to DATA when in_bit == 0.
DATA	Capturing Payload. The FSM enables the shift register to catch the 8 data bits. It increments the counter on each clock edge.	Jumps to PARITY when bit_count == 7
PARITY	Capturing Parity. The FSM pauses shifting and catches the single Even Parity bit into a dedicated register.	Jumps to STOP unconditionally on the next clock cycle.
STOP	Verification and Output. The FSM checks the final Stop Bit and calculates the XOR parity tree. If valid, it pulses done high. If invalid, it pulses frame_err or parity_err.	Jumps to IDLE unconditionally on the next clock cycle to await the next packet.



4. Bonus Challenge: Command Receiver

This is an upgraded version of the Frame Packet Receiver. It includes an additional state to capture a 2-

bit command header before capturing the 8-bit data payload, allowing it to act as a simple ALU.

State Specifications

State	Meaning	Transition condition
IDLE	Waiting for Start Bit.	Jumps to CMD when in_bit == 0.
CMD	Capturing Command. Enables a 2-bit shift register to catch the operation code (LOAD_A, LOAD_B, ADD, or CLEAR).	Jumps to DATA when bit_count == 1.
DATA	Capturing Payload. Enables the 8-bit shift register to catch the data payload.	Jumps to PARITY when bit_count == 7.
PARITY	Capturing Parity. Catches the parity bit.	Jumps to STOP unconditionally.
STOP	Execution. Performs error checks. If no errors are found, it	Jumps to IDLE unconditionally.

	executes the command stored in the CMD register on internal registers A and B, then outputs the result.	
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